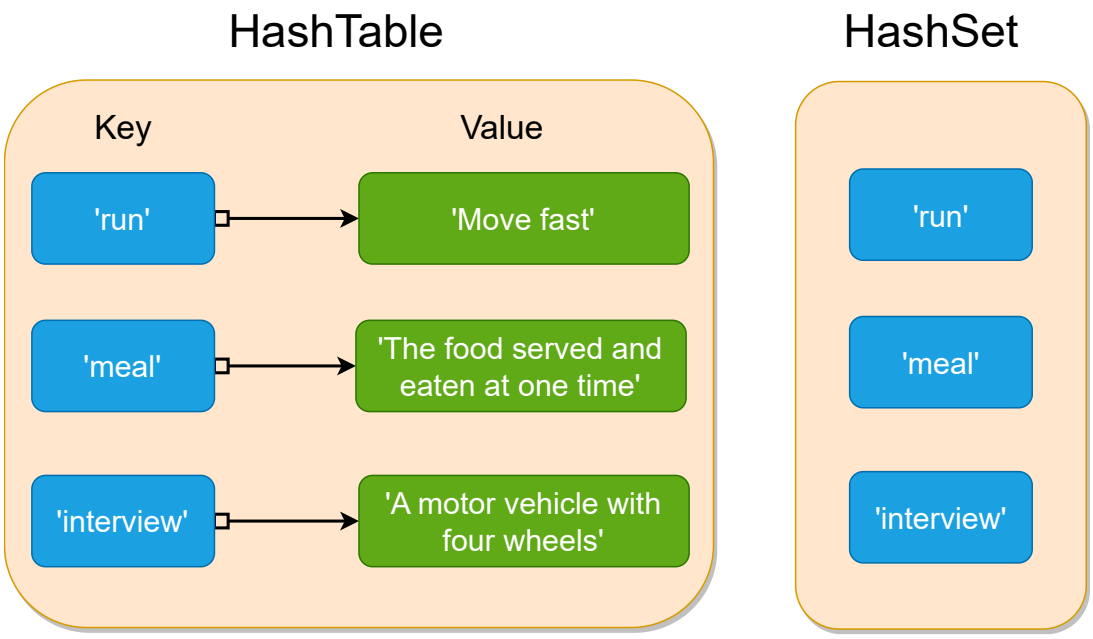
Introduction to HashSets

In simple terms, a HashSet is a collection or a group of unique elements. Think of it as a party where everyone has a different name; no two people can have the same name. In coding terms, this means HashSets do not allow duplicate values.

The Difference at a Glance: HashTable vs. HashSet

HashTable is like a dictionary, full of word (key) and definition (value) pairs. In contrast, HashSet is more like a unique-word collector, not bothered about definitions, just the words.



HashTable vs. HashSet

Core Characteristics of HashSets

- **Uniqueness**: The first thing to remember about HashSets is that every element is unique. It's like having a basket of fruits, and no two fruits in it are the same. This uniqueness is the superpower of HashSets!
- **Null Elements**: A HashSet can include a null element. It's like having an empty slot in your fruit basket, and that's perfectly okay!
- **Order**: When it comes to order, HashSets like to keep it casual. They do not guarantee any specific order of the elements. It's like reaching into your fruit basket blindfolded – you never know which fruit you'll grab first!

Internal Working of a HashSet

- **Handling Duplicates**: So, what happens when HashSets encounter a duplicate? They simply ignore it! If you try to add a fruit that's already in your basket, HashSet will just keep the original and discard the duplicate.
- **Hashing Function**: HashSets, like HashTables, use a hashing function to decide where each element should go. It's like assigning a specific spot in your basket for each fruit based on its name.
- **Resizing the Set**: What happens when your basket gets full? When a HashSet starts running out of room, they increase the basket size to make space for more unique elements.

In the next section, we will solve some problems related to HashSets.

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Solution: Ransom Note (Easy)

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Problem 1: Counting Elements (Easy)

Mark as Completed